

Quiz 12

$\begin{matrix} G & \text{Age} & \text{stu} \\ f & 20 & \text{yes} \end{matrix}$

yes $P(x|G_i) \times P(G_i)$ $P(\text{buys} = \text{yes}) = 9/15 = 0.643$, $P(x|\text{yes}) = \frac{2}{9} \times \frac{4}{9} \times \frac{6}{9} \times \frac{9}{9} = 0.044$
 $\therefore 0.044 \times 0.643 = 0.028$

No $P(x|G_i) \times P(G_i)$ $P(\text{buys} = \text{no}) = 5/14 = 0.357$, $P(x|\text{no}) = \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} = 0.019$
 $\therefore 0.019 \times 0.357 = 0.007$

ans yes with no (0.028 > 0.007)

$\therefore \frac{4}{5}$